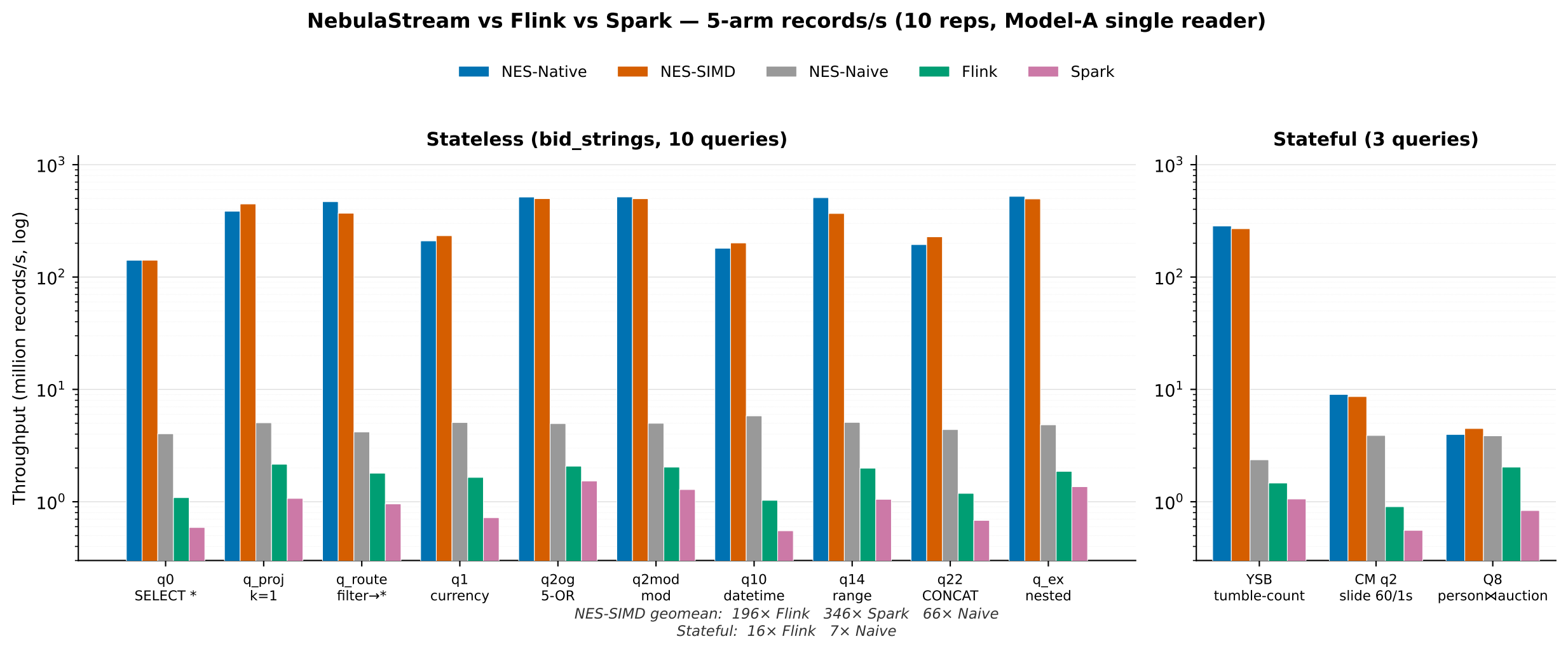
NebulaStream vs Flink vs Spark — 5-arm records/s (10 reps, Model-A single reader)
NES-Native
NES-SIMD
NES-Naive
Flink
Spark
Stateless (bid_strings, 10 queries)
Stateful (3 queries)
Throughput (million records/s, log)
10^3
10^2
10^1
10^0
q0
SELECT *
q_proj
k=1
q_route
filter→*
q1
currency
q2og
5-OR
q2mod
mod
q10
datetime
q14
range
q22
CONCAT
q_ex
nested
YSB
tumble-count
CM q2
slide 60/1s
Q8
person⋈auction
NES-SIMD geomean: 196× Flink 346× Spark 66× Naive
Stateful: 16× Flink 7× Naive